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01TVEQW - Electrical Technologies for eMobility

Final projects

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1. Project goal

The goal of the project is the sizing of a power conversion system for xEV, as shown in Fig. 1.

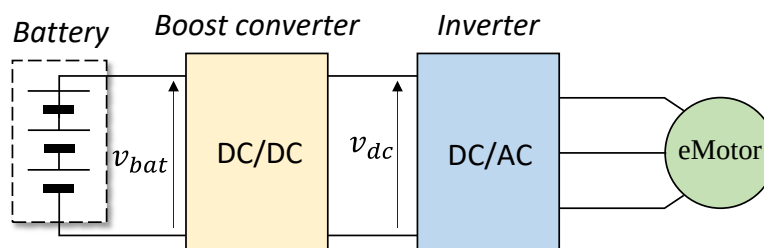


Figure 1. xEV power conversion system.

The considered power conversion system consists of:

- Traction HV battery.
- DC-DC boost converter. For simplicity, the DC-DC converter will include a single bi-directional switching cell, although for high power levels (> 50 kW) the converter may have a modular structure with more cells operating in parallel.

- Three-phase inverter.
- Synchronous electrical motor.

The students will be split in groups of maximum 5 people. Each group will receive a Matlab file with the data of the DC-DC converter and of a real electric motor prototype intended for e-drive applications, together with its operating specifications.

The provided Matlab files contain:

- Rated voltage of traction battery. For simplicity, the battery shall be considered as a constant voltage source.
- DC/DC converter rated output voltage, rated power along with some specifications (switching frequency, input current ripple, output capacitance).
- The flux maps of an electric synchronous motor drive and a typical working cycle of that motor in the torque-speed plane.

All the data are included in a Matlab file named “*DataXX.mat*”, where *XX* is the number of the drive.

The steps that must be performed for the project are:

1. Sizing of the input inductor of the DC-DC converter. Computation of the voltage stress for power devices and define the rated voltage of power devices. Computation of the current stress for the power devices and the input inductor.
2. Design of the DC-DC converter cascaded control.
3. Simulation verification DC-DC converter operation.
4. Retrieving the machine type and fundamental parameters based on the given data.
5. Sizing the inverter based on the motor data and operating cycle.
6. Calibrating a FOC algorithm for the provided motor.
7. Implementing in Simulink the model of the machine and testing the performances of the drive in the operating cycle without the DC-DC converter.
8. **Simulation of the whole system in Simulink for at least one machine operating point, using average dynamic models for DC-DC converter and inverter.**

2. Project deadline and delivery

For each student, the deadline for submitting the project is 7 days before the end of the exam session when you will do the exam:

- If you pass the exam in the Summer session (17/06/24-27/07/24) the project must be delivered at the latest on 20/07/22
- If you pass the exam in the Autumn session (02/09/24-21/09/24) the project must be delivered at the latest on 14/09/22
- For any following exam session, the Polito calendar is not yet defined, but the “7 days” rule will still apply. You can find the exam sessions calendar [here](#).

Once delivered, the project mark is valid for the full academic year, i.e. until the summer session of 2025 excluded. In light of these roles, and considering that the working groups can include up to 5 students, **we recommend to form the working groups aggregating people planning the exam in the same session.**

At the end of the work, each group is expected to **submit the project for evaluation**. For submitting the project, the students are required to send to the teachers an email including the link to a shared folder, using one of the commercial clouds (dropbox, gDrive, OneDrive, ...). The email must contain the group data (drive under test, names and surnames of the group participants). The shared folder must include:

1. A **pdf report** describing the adopted workflow, procedure and equations and supported by results and conclusions. The report should be written according to the provided template. You may add or remove sections from the template if necessary. Despite the carried work must be properly and clearly described, the ability to synthesize will be appreciated, as well as a good quality and clarity of the figures.
2. The **final simulation files** aggregating the DC/DC converter, the inverter, the control and the motor. If any instruction is not automatic, such as the parameters initialization, a ReadMe file with instructions on how to run the simulation should be added.

It is not acceptable submitting the project to the “elaborati” page in the PoliTo portal.

3. Provided data

For each group, the provided data are contained in a Matlab file named “*DataXX.m*”, where XX is the number of the data set. This includes the specifications and characteristics of the DC/DC converter and the e-motor.

3.1. DC-DC converter

As already mentioned above, the provided DC-DC converter data include the rated power (kW), the rated output voltage (V), the battery voltage (V) the switching frequency (kHz), the output capacitance (μF), the input current ripple (%) with respect to the mean rated input current. All these data are stored in the variables “*DCDC_Pmax_kW*”, “*DCDC_vout*”, “*DCDC_vbatt*”, “*DCDC_fsw*”, “*DCDC_Co*”, “*DCDC_dlbatt*”.

3.2. Synchronous motor drive

The provided motor data contain a set of matrices describing the flux maps of the Motor Under Test (MUT). Each file corresponds to a **synchronous machine** of one of the types described during the course (SPM, IPM, PM-SyR, SyR, not the same for different groups). The orientation of dq axes is defined accordingly.

In particular, the file *DataXX.m*, where XX is the number of the analyzed MUT, contains the magnetic model of the machine (flux linkages as a function of the stator currents). The variables I_d and I_q define a regularly spaced grid of points in the dq current plane. For each of those points, the matrices F_d and F_q , where F stands for “flux”, provide the corresponding λ_d and λ_q respectively. As an example, you can visualize the flux maps with the following commands:

```
>> figure
subplot(1,2,1)
mesh(Id,Iq,Fd)
xlabel('$i_d$ (A)'); ylabel('$i_q$ (A)'); zlabel('$\lambda_d$ (Vs)');
view(3)
subplot(1,2,2)
mesh(Id,Iq,Fq)
xlabel('$i_d$ (A)'); ylabel('$i_q$ (A)'); zlabel('$\lambda_q$ (Vs)');
view(3)
```

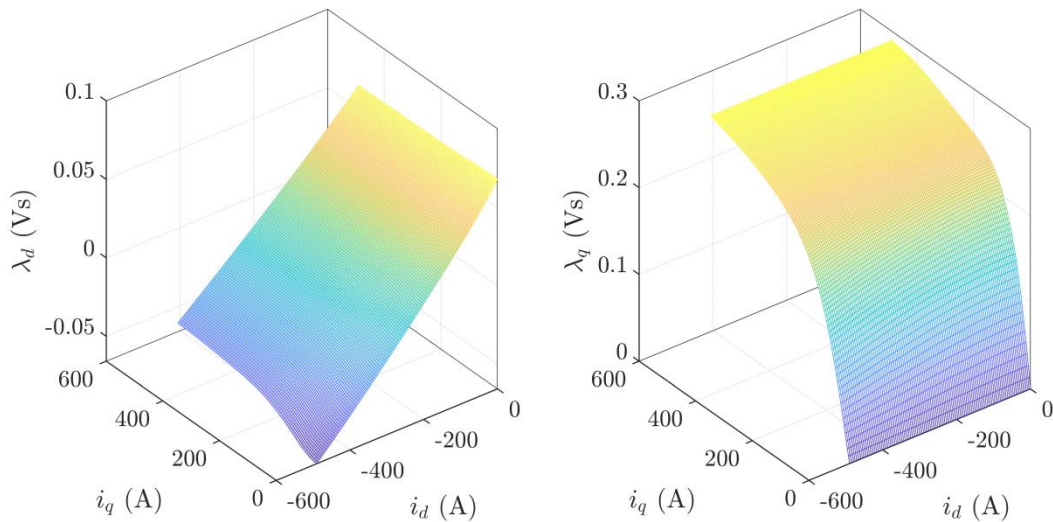


Figure 1. Example of flux maps

Note that **i_d varies along the matrix rows, while i_q varies along the columns**. You can also plot a single row or column for fixing one of the current components. As an example, the Matlab command:

plot(Id(:,1),Fd(:,1))

would plot the characteristic of $\lambda_d(i_d, i_q = i_{q0})$ where i_{q0} is the value of i_q in the first column of the Iq matrix, i.e. the minimum i_q value in the flux maps. You can also access single points of the flux maps or interpolate the values. As an example, the Matlab command:

interp2(Id,Iq,Fd,-5,12)

would interpolate the value of λ_d for $i_d = -5A$ and $i_q = 12A$.

The provided flux maps do not cover the four quadrants of the dq plane. Depending on the machine type, the symmetry of the magnetic characteristic should be exploited to cover the remaining quadrants.

The file “DataXX.mat” also contains additional data that can be used for manipulating the flux maps. **Not the same data are given for each group**. The additional data may include:

- The number of pole pairs p
- The stator resistance R_s

- The MTPA trajectory, described by the vectors id_KtMax , iq_KtMax and the correspondent torque T_KtMax
- The MTPV trajectory, described by the vectors id_KvMax , iq_KvMax and the correspondent torque T_KvMax
- The torque map T , in a regular grid matrix with the same number of points of Id , Iq , Fd , Fq variables

The provided motor data come from experimental measurements of real motor prototypes. Limited deviations from the theoretical MTPA and MTPV trajectories can be expected, as well as eventual noise in the data. As an example, the torque contour of one of the machines is reported here, together with the MTPA and MTPV trajectories.

```
>> figure
[cl,h1] = contour(Id,Iq,T,'r','LevelStep',10);
set(gca,'DataAspectRatio',[1 1 1])
plot(id_KtMax,iq_KtMax,'b')
plot(id_KvMax,iq_KvMax,'k')
legend('T','MTPA','MTPV')
```

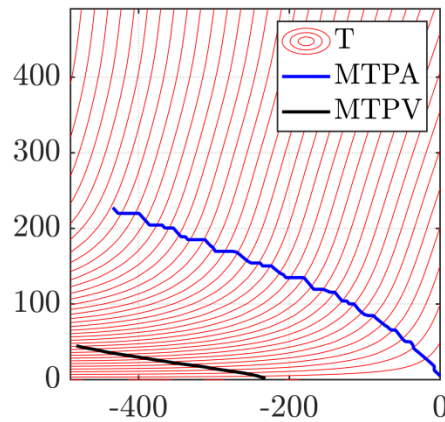


Figure 2. Example of torque contour, MTPA and MTPV.

Finally, a typical operating cycle is given for the considered machine, still in the file *MUTXX.m*, where the vector *Cycle_n* is the mechanical speed in rpm and the vector *Cycle_T* contains the related torque request. An example is given in the following Figure.

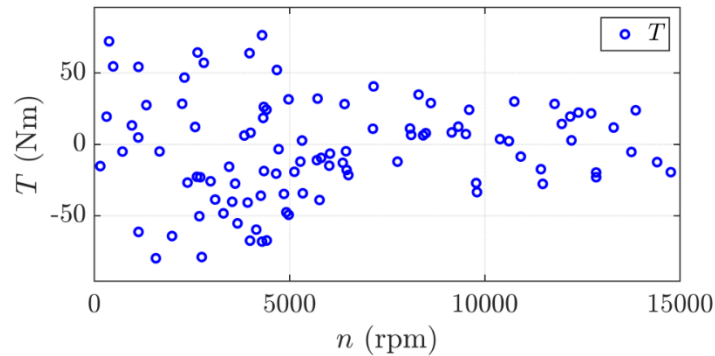


Figure 3. Example of required operating points.

4. Suggested operating procedure

4.1. DC-DC converter

The Exercise 01 – DC-DC converter and Lesson 04 are enough to tackle properly the part related to the DC-DC converter. Some suggestions are presented below, with reference to the steps of the project presented previously:

(1) Consider the DC-DC converter operating point at rated power and rated output voltage, taking into account the provided battery voltage. The input current ripple (peak-to-peak) (%) is considered as:

$$\Delta i_{in,pk-pk}(\%) = \frac{\Delta i_{in,pk-pk}}{I_{in,rated}} \cdot 100$$

The input current must be calculated taking into account the converter efficiency (unknown if the losses cannot be calculated). A good initial value of the efficiency is 95%.

The rated voltage of automotive power devices can be 600V, 650V, 1200V. The currents stress is intended as mean current and RMS current, to be calculated for the rated operating point.

(2) The design of the cascaded control means the calculation of the gains for the current controller and voltage controller, according to the maximum allowed crossover frequencies for the current control loop and the voltage control loop.

(3) It is recommended to use Simulink for the simulation, although PLECS can be used as well (please refer to PLECS manual).

(4) The application of the load current for the DC-DC converter should be done with a slew-rate limiter (using the block Rate Limiter from Simulink/Discontinuities) to reach the final value in no less than 50 ms.

4.2. eMotor

At first, it is required to understand what type of synchronous motor you received. Based on the flux maps, you should understand if it is a SPM, IPM, PM-SyR or SyR.

Tips: is it isotropic or not (differential inductances)? Are there PMs? In what quadrant the MTPA is?

Still based on the flux maps, it is required to identify the rated magnetic parameters of the machine (rated inductances for d and q axes in the linear and saturation regions and PM flux). With this respect, **the magnetic saturation must be considered.**

Tip: plot the apparent and the differential inductances on varying the current. Then, determine a nominal value for L_d, L_q, l_d, l_q . What inductance do you need for motor control simulation? What for FOC calibration?

Then, the inverter must be sized in order to satisfy the given $T - \omega$ operating cycle. The values to be determined are:

- Inverter current rating

- Inverter DC-link voltage v_{dc}
- Minimum switching frequency f_{sw}

The selected parameters must allow a safe operation of the drive under the target working cycle. Anyway, the inverter should not be too much oversized since the inverter losses normally increase with v_{dc} and f_{sw} and its cost is highly affected by the current and voltage rating. A reasonable trade-off should be achieved, e.g. a safety margin of 10%. For this task, assume a good flux weakening strategy is implemented. **The rated machine current (nameplate) should also be determined.**

Tips: what current vector corresponds to the maximum required torque? What is the base speed? What trajectory is followed by the current vector in flux weakening? And what's the correspondent flux and voltage? Remember that the f_{sw} must be sufficiently high for permitting to control the motor at the maximum speed. Be careful with the measurement units!

As an example, the following Figure reports a driving cycle (blue dots) and the operating limits imposed by 2 inverters INV1 and INV2. As can be seen, the size of INV2 is not sufficient for covering all the required operating points.

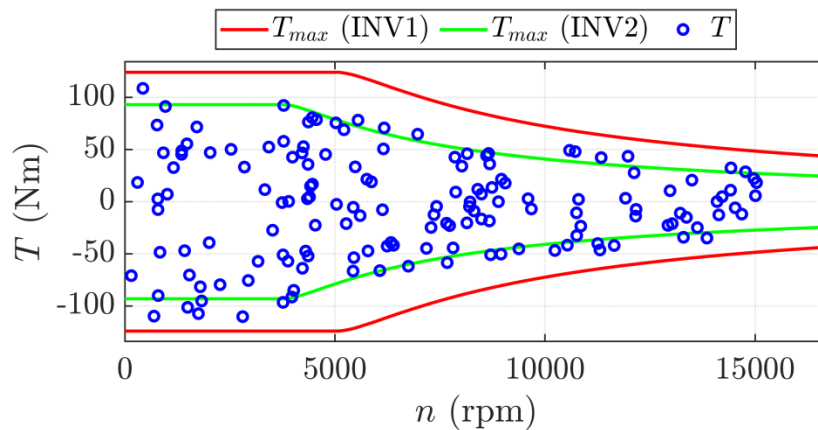


Figure 4. Examples of inverter sizing based on a driving cycle

For retrieving an approximated maximum torque profile of the drive under flux weakening, you can assume a constant maximum torque capability below the base speed. Over the base speed, it is suggested to **rotate the current vector anticlockwise, computing the corresponding flux amplitude and** finally, based on the selected v_{dc} , **retrieve the approximated maximum speed profile** of the drive (what approximation are you introducing?).

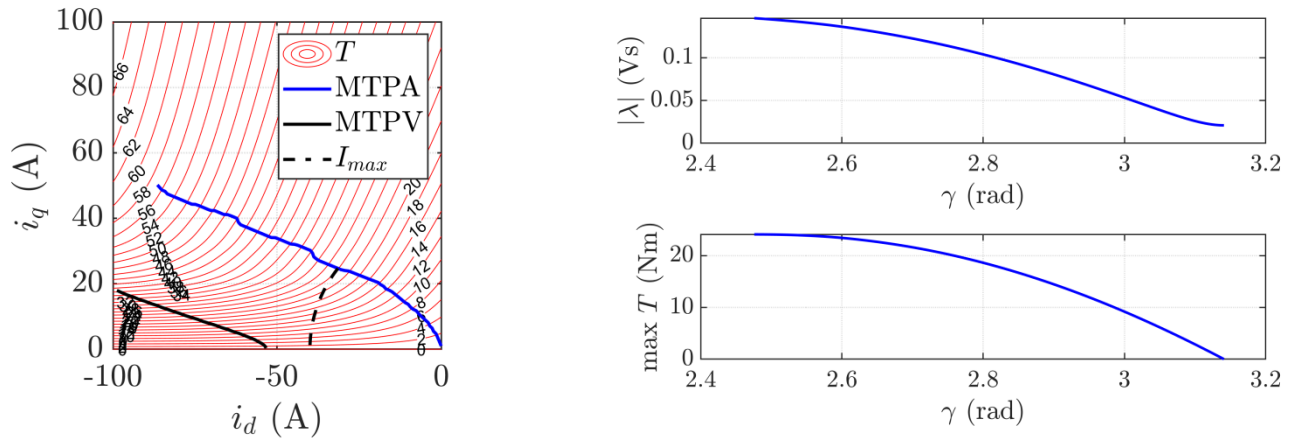


Figure 5. Example of max current profile during FW (left) and correspondent flux amplitude and torque (right).

Tip: the suggested Matlab function is `interp2()`;

Then, a CVC-FOC algorithm must be calibrated. This requires computing a proper value of the proportional and integral regulators in d and q axes, again based on the machine flux maps. For a proper torque control, a high bandwidth is normally desired. Anyway, the control algorithm should be stable regardless the operating region, i.e. also when magnetic saturation is achieved.

Tips: what is the most critical areas in the $T - \omega$ plane for control stability? What inductance do you use?

4.3. Tests in Simulink

03

Finally, the Simulink file provided with the Exercise 04 should be modified to represent the drive under test. The required modifications are:

- Implement the feed-forward of the motional term
- Modify the motor model to represent the machine under test
- Implement the FOC calibration retrieved at the point 4.2.

For this task, a linear magnetic model can be implemented, calibrated according to representative machine inductances. You should test the drive:

- below the base speed, from no-load up to the maximum torque operating on MTPA trajectory.
- one point in flux weakening speed range should be tested by directly setting the current vector reference i_{dq}^* .

Note that testing the torque capability requires to modify the Simulink model by implementing the MTPA law. It is suggested to use two blocks "1-D LookupTable" to read the vector T_KtMax and interpolate the corresponding i_{d_KtMax} and i_{q_KtMax} . The figures below depict the suggested Simulink implementation and expected results. You can start from the Simulink model provided for Exercise 4, adapt the motor parameters declaration and PI calibration and, if the machine is not isotropic, split the d and q axes in the model.

Tips: For testing single operating points, you can manually set the required i_d^* and i_q^* . For testing the torque capability below the base speed, set a ramp torque reference from zero to the maximum torque and exploit the MTPA LUTs. For the point in FW, manually bypass the MTPA LUTs and directly set the proper i_d^* and i_q^* based on the flux maps and torque contours.

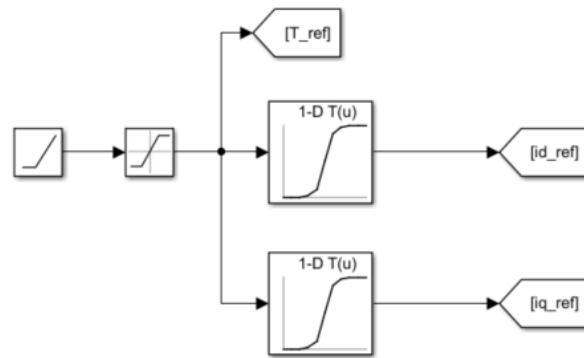


Figure 6. Simulink implementation of MTPA low, to be used below base speed.

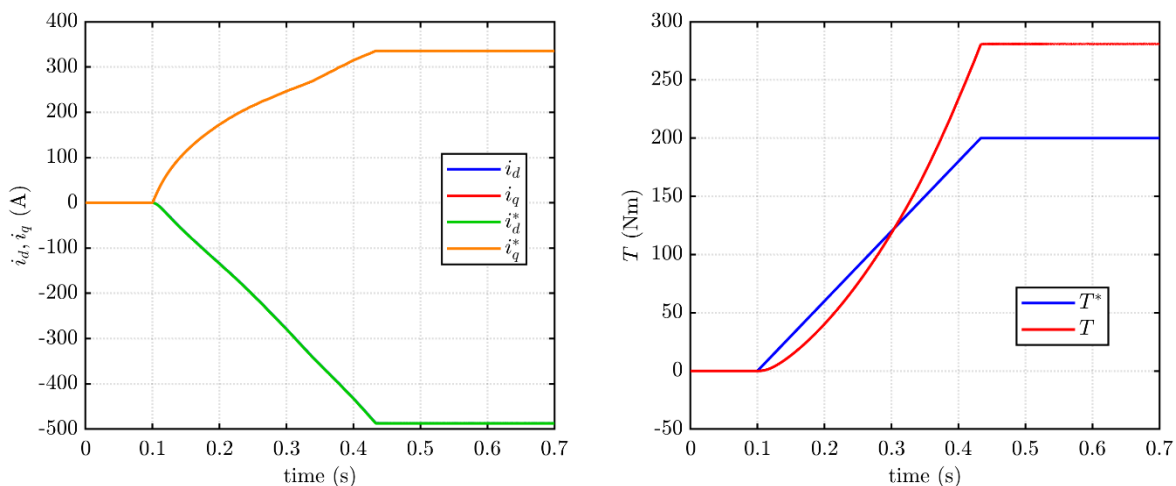


Figure 7. Simulation results for a torque reference ramp below the base speed. a) current reference and measured; b) torque reference and measured.

Because for computing the current I get the LUT why for the motor model I use constant inductance that does not consider saturation effect

NOTE: the obtained torque is not equal with the reference. Why? How could the problem be solved?

4.4. Full system simulation

Finally, the full system should be simulated, including the DC/DC and the motor drive. For this task, you can merge the Simulink models obtained at the Sections 4.1 and 4.3. It is required to simulate at least one point below the base speed.